

Wavelength-dependent differential excitation of cobalt ferrite and implications for Raman spectroscopy structural quality control

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INTRODUCTION

Cobalt ferrite (Co-ferrite) is one of the most versatile structure in the ferrite family of nanomaterials. It can be substituted with a plethora of elements, and lattice defects can be artificially induced to achieve unique properties. In addition to its traditional and emerging applications such as medicine, catalysis, energy storage and memory devices, spin-tronics and nanorobotics (Ansari et al., 2025), Co-ferrite also spontaneous Faraday rotation, a property that qualifies it for promising potential applications in opto-electronics and magneto-optical devices. For instance, optical isolators are critical components that protect sensitive electronics from back-reflection. Conventional designs are often difficult to produce and cannot be miniaturized due to the requirement of the permanent presence of a source of magnetic field. Recent studies have achieved Co-ferrite optical insulators in a compact format suitable for integration into optical waveguides (Sato et al., 2024; Serrano-Núñez et al., 2022)

Furthermore, Co-ferrite nanoparticles complexed with polymers to achieve core-shell structures, further embedded into a polymer matrix which show favourable properties for innovative magneto-optic sensor (Lopez-Santiago et al., 2012). Furthermore, a magneto-optical wave guide was achieved using Co-ferrite NPs embedded in a silica-zirconium matrix (Choueikani et al., 2009). The electronic absorption maximum of Co-ferrite red-shifts with increasing particle size (Gharibshahi et al., 2020), as is common for nanomaterials. Ferrites generally exhibit excellent UV-blocking capabilities, making them suitable for applications in UV detectors and photodiodes, with Co-ferrite being a strong absorber in the range $190 - 2000 \text{ cm}^{-1}$ (El-Masry and Arman, 2025)mparative characterization of common ferrites Co-ferrite, nickel ferrite and zinc ferrite, including their optical absorption, band gap energies, Urbach energy, refractive indices and optical conductivity. Due to visible range absorption and photoluminescence of surface defects, Co-ferrite NPs were also considered as magnetically recoverable photocatalysts (Selima et al., 2021).

A sub-field of magnetoplasmonics seeks to achieve colloidal nanoparticles comprising a component such as cobalt ferrite, and a noble metal plasmonic component. This type of substrate would be extremely useful in high-precision analytics, combining the advantages of plasmonic enhancement of optical signals of adsorbed analyte and the ability to dynamically reconstruct the substrate by application of external magnetic fields. Several synthesis methods have been developed for reproducible favrication of Co-ferrite NPs such as coprecipitation, various chemical and physical methods (Baraliya and Joshi, 2014; Lazarevic et al., 2024) nd thin films such as pulse laser deposition, spin-coating, RF-deposition and many more (Foerster et al., 2012; López-Moreno et al., 2022), with fien tunability of fabrication parameters well explored. The synthesis prospects of colloidal nanoparticles combining Co-ferrite and gold/silver in various core-shell and decorated

particles have been well explored (REFS). However, a knowledge gap persists concerning the underlying excitation processes of Co-ferrite under different laser wavelengths. Namely, exploitation of the silver LSPR requires excitation in green spectral range (400 – 500 nm), while in gold the LSPR is active in red to NIR range (600 – 800 nm). The behaviour of the Co-ferrite component under such excitation, and could it experimentally influence the plasmonic component, was hitherto not systematically studied.

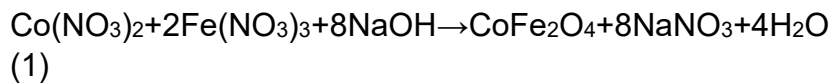
Magnetite, Mn-spinels and Mn-Zn ferrites are commonly analysed under red wavelength range (632.8 to 647.1 nm) as they rapidly transition to defectuous and oxidized maghemite and hematite phase under relatively low laser energy input threshold causing local heating (Buciuman et al., 1999; Chourpa et al., 2005; Nekvapil et al., 2020). Co-ferrites are commonly analysed using laser energy in green wavelength region (488 to 532 nm) (Baraliya and Joshi, 2014; Chandramohan et al., 2011; Dhaka et al., 2021; Foerster et al., 2012; Lazarevic et al., 2024; López-Moreno et al., 2022; Saccone et al., 2015; Yadav et al., 2017), as they seem to be more thermally stable, and hence this laser line can be considered as baseline. Co-ferrites were occasionally analysed under red laser excitation (Bui et al., 2014), however, in the context of nanomaterials engineering and thus lacking a comprehensive treatment of spectra. UV lasers are rarely used, except for exceptionally stable Zn-ferrite (Albini et al., 2023), while NIR lasers have also been used on some occasions (A Flores et al., 2020; Thanh et al., 2024). However, the spectra in this case often require heavy processing due to low scattering efficiency under longer wavelength radiation. Bands configuration in Raman spectra of Co-Zn ferrites recorded under 632.8 nm excitation (Szatmari et al., 2022) seem slightly unusual, in the sense that the sum of spectral contributions of either Co- or Zn- related bands seem too large relative to Fe-related bands. On another note, a wavelength-dependent enhancement of the LO phonons was observed in zinc oxide (ZnO) nanoparticles synthesized by different methods. Here, these phonons were relatively weak under the non-resonant 532 nm laser line, however, exhibited notable enhancement under the near-resonant 442 nm line, causing apparent significant shifting and broadening of otherwise sharp bands (Nekvapil et al., 2022). Excitation of ZnO with even shorter wavelengths closer to the resonance maximum activates up to 6th overtone of the LO phonon (Zhu et al., 2009).

In this study, we explore in depth comparatively the Raman response of Co-ferrite NPs under 442, 632.8 and 830.8 nm excitation lines, as opposed to the commonly used 532 nm line. We aim to identify which type of information about the material structure can be revealed by less common excitation lines in a comprehensive Raman spectroscopy study. We report on the wavelength-dependent enhancement of the Co-related sublattice under the latter excitation line and its implications for sample characterization strategy.

2. MATERIALS AND METHODS

2.1. Ferrite synthesis procedures

CoFe₂O₄ (CF) nanoparticles were synthesized via a co-precipitation method, following a previously reported procedure with slight modifications to suit the objectives of the present study (Stefan et al., 2021). Iron nitrate and cobalt nitrate hexahydrate were dissolved in 100 mL of Milli-Q water at a molar ratio of 2:1. The resulting solution was ultrasonicated for 30 min to ensure complete homogenization. A 4 M NaOH solution was then added dropwise under continuous stirring to adjust the pH to 12, leading to the formation of a brown precipitate. The overall reaction can be described as follows (eq 1):



The mixture was maintained under magnetic stirring at 80 °C and 900 rpm for 2 h. The precipitate was washed several times with double-distilled water until neutral pH was achieved and subsequently dried at 70 °C for 12 h. Finally, the dried powder was calcined at 600 °C for 2 h to improve crystallinity and phase purity.

Table 1. List of Co-ferrite samples prepared along with their lattice parameter (*a*) and crystallite size (*d*) determined by XRD, and structure as determined by Raman spectroscopy.

Sample code	Synthesis temperature (°C)	Annealing temperature (°C)	<i>a</i> (Å)	<i>d</i> (nm)	Structure formula A[B]O ₄
CoF-P	80	n/a	8.32(1)	11.1	Co _{0.55} Fe _{0.45} [Co _{0.45} Fe _{1.55}]O ₄
CoF-200	80	200	8.34(1)	11.3	Co _{0.52} Fe _{0.48} [Co _{0.48} Fe _{1.52}]O ₄
CoF-300	80	300	8.36(1)	14.5	Co _{0.48} Fe _{0.52} [Co _{0.52} Fe _{1.48}]O ₄
CoF-400	80	400	8.39(1)	13	Co _{0.52} Fe _{0.48} [Co _{0.48} Fe _{1.52}]O ₄
CoF-500	80	500	8.37(1)	14	Co _{0.5} Fe _{0.5} [Co _{0.5} Fe _{1.5}]O ₄
CoF-600	80	600	8.38(1)	15	Co _{0.51} Fe _{0.49} [Co _{0.49} Fe _{1.51}]O ₄

Nanoparticles were subsequently annealed to produce a series of samples with slightly different size. Annealing was done separately for 2 hours at temperature

steps of 200, 300, 400, 500 and 600 C, in argon flow, at heating ramp of 10 C/min, finished by abrupt cooling within the tube outside of the oven.

2.2. XRD analysis

X-Ray diffraction patterns were measured using a Bruker D8 Advance diffractometer, equipped with a Cu K α source. The structure and microstructure of the samples was investigated by Rietveld refinement using the Fullprof suite (Rodríguez-Carvajal et al., 2025; Roisnel and Rodríguez-Carvajal, 2001).

2.3. Raman spectroscopy analysis

Raman spectra were recorded on a Renishaw micro-spectrometer, model InVia Reflex, coupled with a Leica microscope, recently upgraded with a new generation of Peltier-cooled detector with improved quantum efficiency. The nominal resolution of the spectrometer is 0.5 cm⁻¹ in NIR and visible range, and 1 cm⁻¹ in the UV range. Four laser lines were used: a He-Cd laser line at 442 nm, a Cobolt DPSS laser emitting at 532 nm, a He-Ne laser emitting at 632.8 nm, and a diode laser emitting at 830 nm. A 100x objective (NA 0.9) was used in all cases, however, spectra acquisition parameters had to be adjusted for each laser line. In either case, care was taken to obtain the highest quality spectra possible (low signal-to-noise ratio), but without damaging the Co-ferrite phase (absence of sharp peaks in <450 cm⁻¹ range). Hence, the laser power was below 20 mW in all cases, while the integration time and number of scans per spectrum were optimised.

At least 5 point spectra were acquired from different locations of each sample under either laser line. Batches of homologous spectra were averaged to facilitate handling of data. Analysis of at least 5 randomly selected point spectra from each sample has shown that there were no outliers that would invalidate the results discussed further on. Spectra were deconvoluted by Lorentzian functions in whole range in the range 70 – 850 cm⁻¹. Such processing method was shown to be relevant for this type of materials with predominantly ionic interactions, and the Lorentzian shapes are commonly for spectra of ferrites. The number of components (peaks) was determined by visual assessment and previous literature reports. Modes referring to A-sites contain “(A)” in their designation, while those referring to B-sites contain “(B)”. Cation vacancies hinting at the maghemite impurity phase and usually occurring in 710 - 730 cm⁻¹ range (Chourpa et al., 2005; Nekvapil et al., 2023, 2020), were not observed in the current samples.

3. RESULTS

3.1. Structure of the samples

XRD and Raman spectroscopy confirmed that the identity of samples as Co-ferrite. No impurities nor trace phases were detected in pristine samples and those treated up to 500 °C, however, trace amounts of FeO were found in the sample annealed at 600 °C, as revealed by XRD. However, Raman spectra were devoid of FeO signals, which is in line with conclusions of our previous study which established that the latter method is not as sensitive to small amounts of non-target phase as the former (Nekvapil et al., 2023). The crystallite size (d), lattice parameter (a) and strain were calculated from the refinement of the diffraction patterns and are shown in Table 1.

The samples show a growth tendency of both the a and the d as the treatment temperature increases. Differential increment rate of a and d can be extracted (Figure S1). from data published for other Co-ferrites (Dhaka et al., 2021; El-Okr et al., 2011; Yadav et al., 2017a), Zn-ferrite (Yadav et al., 2017b), Ni-ferrite (Kamble et al., 2015) and Cu-ferrite (Yadav et al., 2016),. The two main processes mediating size growth of NPs are particles coalescence and Ostwald ripening, both of which may start at few hundred degrees, depending on the nanoparticle type (Chen et al., 2013; Liu et al., 2023).

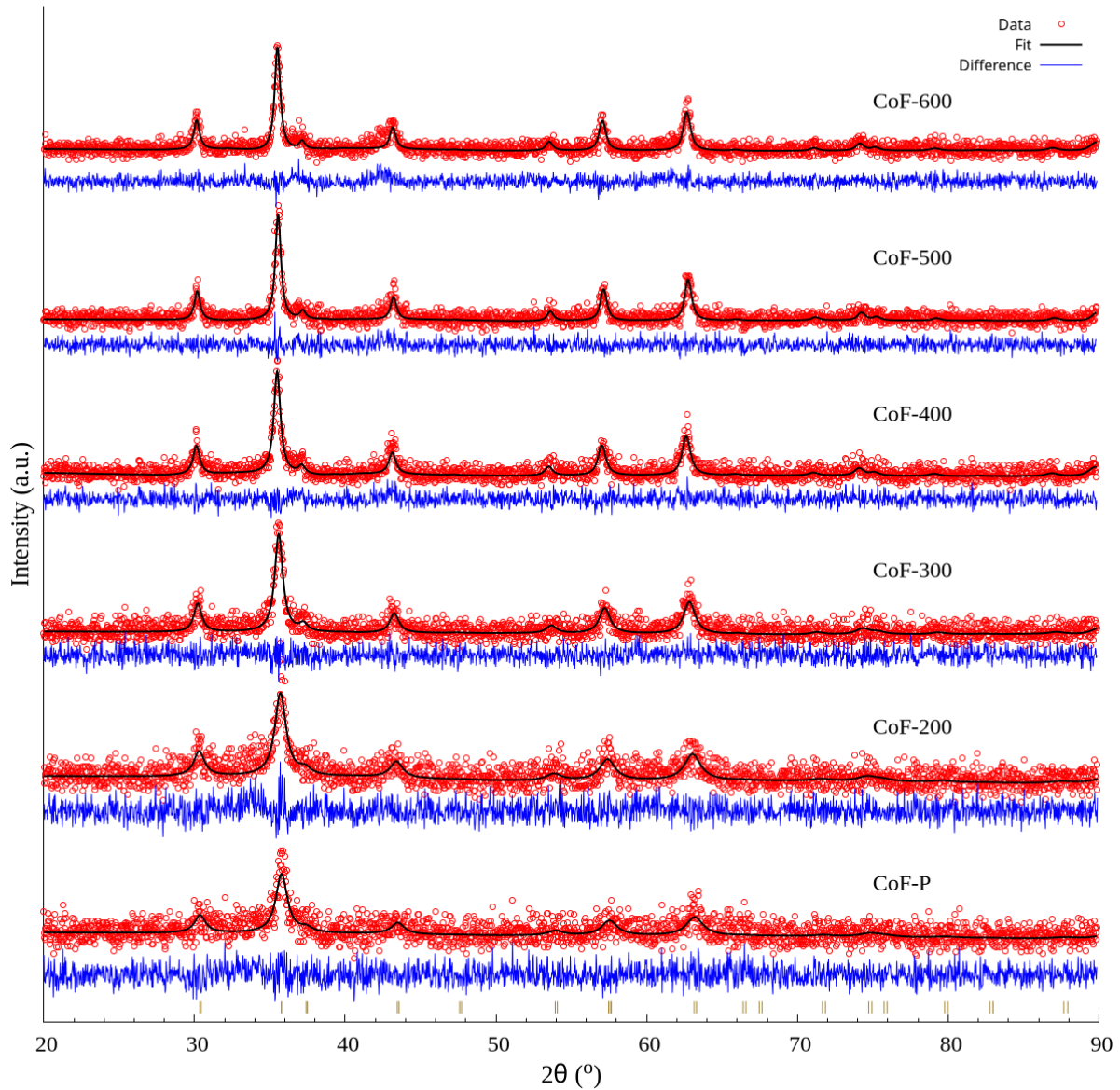


Figure 1. Rietveld refinement results on X-Ray diffraction patterns, for Co-Ferrite samples annealed up to 600 °C. Peak positions are only given once at the bottom of the figure.

The result of the Rietveld refinement of the recorded XRD patterns are given in Figure 1. From this figure we can see that all samples fitted well with the Co-Ferrite structure (Crystallography Open Database, entry 1533163) space group $Fd\bar{3}m$ setting 2. All samples were found to be single phase. Of course, we see that with increasing temperature, the signal to noise ratio increases as well, which is expected due to crystallite size growth and strain removal owing to the annealing process.

Raman spectra of all samples feature the Raman-active modes $T_{2g}(1)$, the E_g , the $T_{2g}(2)$, $T_{2g}(3)$ and the A_{1g} bands in the 100 – 800 cm^{-1} spectral region (Figure 2), as usually reported for Co-ferrites synthesized by different methods (Baraliya and Joshi, 2014; Lazarevic et al., 2024). Each of these bands will be discussed below in

greater detail. The recorded and fitted Raman modes present certain inter-sample variation of their position. Certain amount of this variation can be ascribed to the spin state of Co-ferrite(Almutairi, 2025), however, since the NPs synthesized by this method usually yield ferrimagnetic Co-ferrite, the issue of spin is of minor importance here. Theoretical studies claim that all bands arise exclusively (Shebanova and Lazor, 2003), or at least have contributions from A-sites (Almutairi, 2025)(Almutari). Based on our team's previous experience, we adopt in this paper the approach leaning on experimental observations that the $T_{2g}(2)$ band comprises significant contributions from the B-sites, featuring contributions from both Fe-O(B) and Co-O(B). Furthermore, it seems that there are several naming schemes for Raman-active modes in spinels found across literature. Here, we adopt the naming convention based on symmetry considerations used in previous authoritative theoretical studies (Almutairi, 2025; Shebanova and Lazor, 2003). The mean values for centre and FWHM of all bands recorded under the 532 nm and the 632.8 nm excitation are presented in Table 2 and Table 3, respectively. Modes position and width can be used in future work to assess the appropriateness of the fitting procedure, however, the exact intensities of bands will depend on the particular structure of newly synthesized samples.

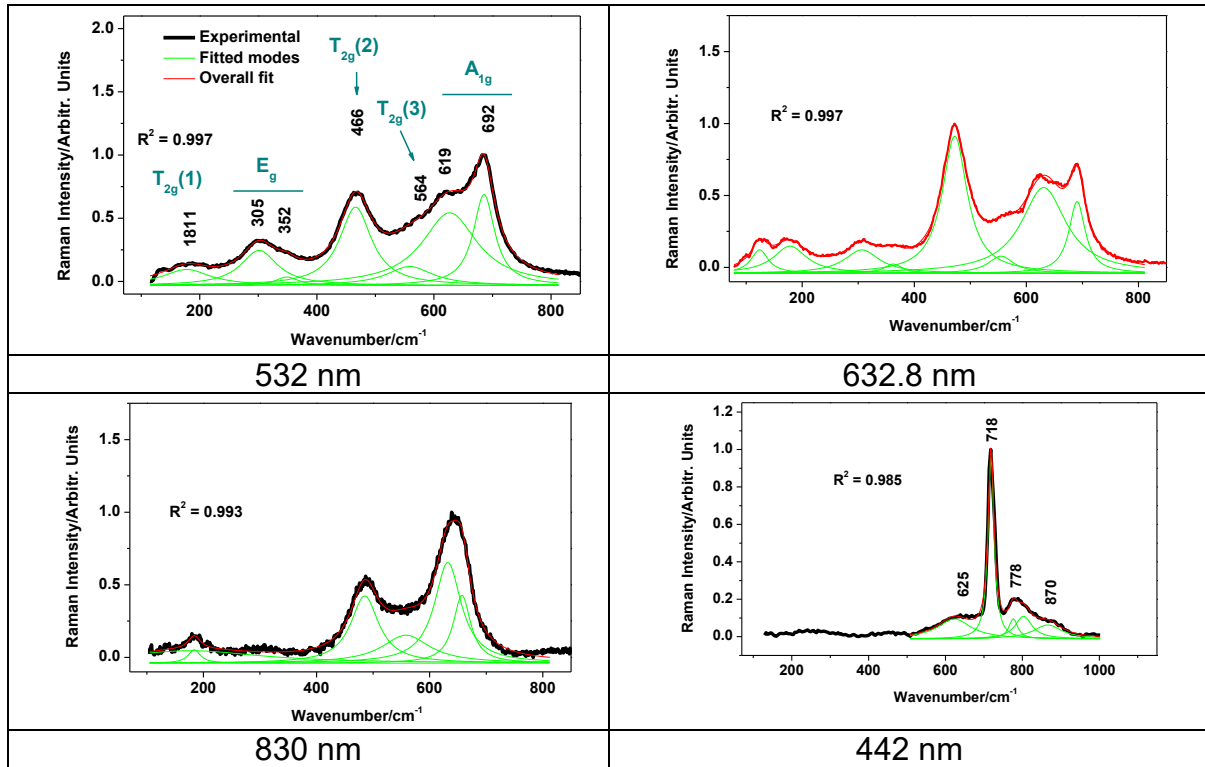


Figure 2. Averaged Raman spectra ($n > 5$) acquired from sample CoF-P under (a) 532, (b) 632.8, (c) 830 and (d) 442 nm excitation. Fitted modes are also shown, the properties of which, including position, FWHM, and normalized intensity and area, are given in Tables 1 – 3. Spectra are background-subtracted and normalized to the strongest signal.

	Mean $\pm$ S.D.			
Mode	Center	FWHM	Area	Intensity
CoF-P				
A _{1g} Fe-O(A)	685	44,18	52,91	0,76
A _{1g} Co-O(A)	628	85.57	64,60	0,48
T _{2g} (3)	564	71.00	31,55	0,20
T _{2g} (2)	467	52.46	54,29	0,57
E _g (b)	352	33.82	3,60	0,06
E _g (a)	305	53.38	26,32	0,23
T _{2g} (1)	182	63.20	5,61	0,06
CoF-200				
A _{1g} Fe-O(A)	684	41.94	53,49	0,81
A _{1g} Co-O(A)	617	84.27	58,32	0,44
T _{2g} (3)	562	67.04	9,82	0,10
T _{2g} (2)	466	52.00	49,38	0,59
E _g (b)	353	28.00	3,02	0,05
E _g (a)	305	56.05	18,449	0,22
T _{2g} (1)	179	74.07	9,11	0,06
CoF-300				
A _{1g} Fe-O(A)	686	40.67	52,42	0,82
A _{1g} Co-O(A)	623	82.95	48,26	0,44
T _{2g} (3)	564	72.35	14,19	0,12
T _{2g} (2)	465	51.00	50,04	0,58
E _g (b)	352	30.72	3,28	0,06
E _g (a)	305	54.55	16,82	0,20
T _{2g} (1)	179	72.81	6,12	0,06
CoF-400				
A _{1g} Fe-O(A)	686	38.92	52,78	0,86
A _{1g} Co-O(A)	618	84.16	56,74	0,42
T _{2g} (3)	564	87.18	7,17	0,08
T _{2g} (2)	467	57.99	46,53	0,60
E _g (b)	351	39.32	2,48	0,05
E _g (a)	304	61.30	18,04	0,22
T _{2g} (1)	181	68.01	6,60	0,06
CoF-500				
A _{1g} Fe-O(A)	685 685	41.85	57,84	0,87
A _{1g} Co-O(A)	617	82.17	57,31	0,44
T _{2g} (3)	562	82.33	8,38	0,10
T _{2g} (2)	465	56.94	48,72	0,61
E _g (b)	352	38.32	2,08	0,05
E _g (a)	304	61.97	19,04	0,22
T _{2g} (1)	180	94.59	10,98	0,07

CoF-600				
A _{1g} Fe-O(A)	686	38.84	52,10	0,85
A _{1g} Co-O(A)	620	80.49	54,29	0,42
T _{2g} (3)	568	113.14	9,34	0,10
T _{2g} (2)	466	63.97	47,14	0,59
E _g (b)	347	40.03	2,79	0,06
E _g (a)	301	73.17	15,54	0,20
T _{2g} (1)	182	69.90	5,35	0,05

632.8

	Mean ± S.D.			
Mode	Center	FWHM	Area	Intensity
CoF-P				
A _{1g} Fe-O(A)	690	35.81	27,73	0,50
A _{1g} Co-O(A)	630	91.87	81,58	0,57
T _{2g} (3)	564.10	82.78	9,64	0,11
T _{2g} (2)	472.19	51.75	72,63	0,91
E _g (b)	359	36.82	2,31	0,04
E _g (a)	306	61.43	8,21	0,11
T _{2g} (1)a	178	61.17	11,26	0,140,11
T _{2g} (1)b	125	35.35	6,65	0,13
CoF-200				
A _{1g} Fe-O(A)	690	33.47	26,89	0,51
A _{1g} Co-O(A)	631	88,51	80,42	0,58
T _{2g} (3)	563.35	71.58	9,57	0,12
T _{2g} (2)	472.75	49.93	74,95	0,94
E _g (b)	358	38.76	3,92	0,06
E _g (a)	306	60.79	15,13	0,13
T _{2g} (1)a	178	50.40	15,64	0,15
T _{2g} (1)b	126	29.31	6,30	0,11
CoF-300				
A _{1g} Fe-O(A)	690	34.80	29,33	0,54
A _{1g} Co-O(A)	631	95.29	84,30	0,56
T _{2g} (3)	565.55	76.97	9,92	0,13
T _{2g} (2)	472.37	53.25	75,77	0,91
E _g (b)	363	43.21	3,34	0,05
E _g (a)	308	73.60	16,54	0,14
T _{2g} (1)a	177	61.45	18,41	0,17
T _{2g} (1)b	126	28.03	5,65	0,13
CoF-400				
A _{1g} Fe-O(A)	690	33.26	29,34	0,56
A _{1g} Co-O(A)	632	101.19	82,57	0,52
T _{2g} (3)	564.56	65.67	9,32	0,13
T _{2g} (2)	472.40	49.20	73,52	0,94

E _g (b)	352	34.93	3,33	0,05
E _g (a)	307	43.38	17,32	0,13
T _{2g} (1)a	178	38.61	10,66	0,13
T _{2g} (1)b	129	21.13	4,26	0,09
CoF-500				
A _{1g} Fe-O(A)	689	35.44	31,12	0,56
A _{1g} Co-O(A)	626	91.02	73,91	0,52
T _{2g} (3)	561.73	61.51	8,16	0,12
T _{2g} (2)	470.75	47.46	74,13	0,96
E _g (b)	355	34.85	4,09	0,07
E _g (a)	307	50.00	14,95	0,15
T _{2g} (1)a	176	43.91	14,27	0,14
T _{2g} (1)b	127	21.80	4,45	0,10
CoF-600				
A _{1g} Fe-O(A)	689	34.18	30,38	0,56
A _{1g} Co-O(A)	627	89.91	68,72	0,51
T _{2g} (3)	562.98	61.11	6,85	0,11
T _{2g} (2)	471	46.39	49,08	0,94
E _g (b)	357	32.94	2,99	0,06
E _g (a)	307	49.98	12,73	0,14
T _{2g} (1)a	176	44.71	12,90	0,14
T _{2g} (1)b	127	25.81	5,58	0,11

830

	Mean ± S.D.			
Mode	Center	FWHM	Area	Intensity
CoF-P				
A _{1g} Fe-O(A)	678	36.78	21.82	0.38
A _{1g} Co-O(A)	634	69.79	66.94	0.64
T _{2g} (3)	547	80.17	60.09	0.50
T _{2g} (2)	471	46.42	62.67	0.88
E _g	299	26.08	5.30	0.13
T _{2g} (1)a	178	36.80	6.09	0.13
T _{2g} (1)b	131	27.12	2.77	0.08
CoF-200				
A _{1g} Fe-O(A)	651	40.20	40.20	0.50
A _{1g} Co-O(A)	626	58.13	50.87	0.57
T _{2g} (3)	552	38.96	26.28	0.2
T _{2g} (2)	483	52.84	43.79	0.54
E _g	2999	20.68	0.97	0.03
T _{2g} (1)a	181	40.96	6.26	0.11
T _{2g} (1)b	132	21.25	1.26	0.04
CoF-300				
A _{1g} Fe-O(A)	683	38.49	18.73	0.31

A _{1g} Co-O(A)	624	84.76	116.19	0.90
T _{2g} (3)	539	71.73	23.11	0.21
T _{2g} (2)	471	52.57	68.31	0.86
E _g	297	27.28	3.77	0.09
T _{2g} (1)a	176	46.87	9.86	0.15
T _{2g} (1)b	128	19.51	1.37	0.05
CoF-400				
A _{1g} Fe-O(A)	6884	33.99	18.67	0.36
A _{1g} Co-O(A)	640	68.19	64.15	0.62
T _{2g} (3)	557	75.15	31.58	0.28
T _{2g} (2)	470	47.48	69.93	0.97
E _g	297	29.13	6.11	0.14
T _{2g} (1)a	182	43.96	8.95	0.15
T _{2g} (1)b	130	28.30	2.41	0.06
CoF-500				
A _{1g} Fe-O(A)	662	56.26	63.44	0.74
A _{1g} Co-O(A)	616	62.99	35.52	0.37
T _{2g} (3)	559	63.33	22.377	0.23
T _{2g} (2)	474	54.09	73.99	0.90
E _g	307	105.64	21.10	0.15
T _{2g} (1)a	187	46.81	9.62	0.14
T _{2g} (1)b	126	45.76	3.07	0.05
CoF-600				
A _{1g} Fe-O(A)	684	33.99	18.667	0.35
A _{1g} Co-O(A)	640	68.19	64.15	0.42
T _{2g} (3)	557	775.15	31.85	0.25
T _{2g} (2)	469	47.49	69.93	0.97
E _g	297	29.13	6.11	0.14
T _{2g} (1)a	182	43.96	8.95	0.14
T _{2g} (1)b	130	28.30	2.42	0.06

There is a consensus both in theoretical and experimental studies that the A_{1g} arises from the vibrations from the A-sites, more specifically A-O stretching and A-O-B bending, and that the band is evidently split into as many components as there are types of tetrahedrally coordinated cations (Almutairi, 2025; Chandramohan et al., 2011). The Fe-O(A) mode consistently centred around 685 cm⁻¹ with common variations of less than 2 cm⁻¹ across the temperature treatments (Figure 3). The Co-O(A) mode exhibited slightly higher variation among temperature steps from 617 to

628 cm^{-1} , however, without any particular trend (Figure 3). In our samples, under the 532 nm excitation. The width (FWHM) of both components decreased with increasing annealing temperature. Narrower Raman bands usually indicate greater vibrational coherence and hence higher crystallinity. Previous reports indicate that the bandwidth of the A-site complex can vary greatly depending on the synthesis method (Baraliya and Joshi, 2014), hence, the somewhat larger FWHM of the Co-O(A) is still in line with bandwidth orders reported for Co-ferrite synthesized by the coprecipitation method. On the other hand, the position of neither mode showed a clear shifting trend across the temperature range.

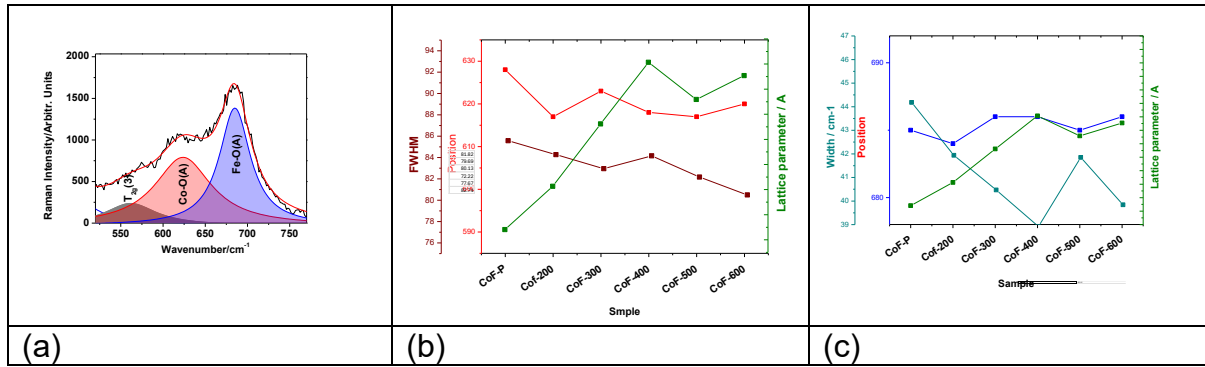


Figure 3. Comparative display of A_{1g} components properties trend across the annealing treatments; (a) zoom of the A_{1g} band highlighting the Fe-O(A) (blue) and the Co-O(A) component (red), alongside the T_{2g}(3) band (grey); position and FWHM evolution of (b) the Co-O(A) mode and (c) the Fe-O(A) mode.

The area fraction of A_{1g} components is indicative of A-site occupancy rate in ferrites, i.e. the inversion degree, termed X (Chandramohan et al., 2011; López-Moreno et al., 2022; Szatmari et al., 2022). According to their lattice energy minimization, Co²⁺ ions are expected to preferentially occupy octahedral sites, while Fe³⁺ are expected to occupy tetrahedral sites (Kovtunenkov, 1997). The inversion degree can be calculated for tetrahedral sites according to area contribution of either mode to the total A_{1g} area. Hence, X_A for Fe^{2,3+} cations can be expressed as $X_{A(\text{Fe})} = \text{area}_{\text{Fe-O(A)}} / \text{area}_{\text{A}_{1g}}$. We have observed a slight increase in $X_{A(\text{Fe})}$ from 0.45 in CoF-P to 0.49 in CoF-600, and up to 0.50 in CoF-500, indicating that some, but not significant, exchange of cations between A and B sites did occur across annealing treatments. Hence, the structure of the sample CoF-P is $\text{Co}_{0.55}\text{Fe}_{0.45}[\text{Co}_{0.45}\text{Fe}_{1.55}]\text{O}_4$, where the section outside the brackets refers to A-sites, and the section inside the brackets to B-sites. The averaged structures were given for all samples earlier in Table 1.

There is experimental evidence that the T_{2g}(2) band arises, at least in big part, from vibrations at octahedral sites, and contains separate contributions attributable to each cation found on this site, homologous to the A_{1g} band components (Datt and Abhyankar, 2017; Himcinschi et al., 2013; Mohammedi and Jali, 2025; Szatmari et al., 2022; Thanh et al., 2024). In some of our spectra, this band could be decomposed into

in some cases a Co-O(B) mode at $452.66 \pm 4.18 \text{ cm}^{-1}$ and a Fe-O(B) mode at $466 \pm 0.89 \text{ cm}^{-1}$. Figure 4 shows such an example. In this example, the total area under Fe-O(A) + Fe-O(B) modes was 1.46 times greater than the total area under Co-O(A) + Co-(B) modes, which is roughly consistent with iron-to-cobalt stoichiometry of the sample, although slightly overweight on the cobalt side. Indeed, spectra of annealed Fe-cobaltite thin films presented by Bui et al. (Bui et al., 2014) show that a Co-rich sublattice may exist alongside the Fe-lattice where Co ions are diffusely distributed. Although that paper follows a different band nomenclature scheme, it is evident in the presented spectra that the Co-rich spinel exhibits the band around 450 cm^{-1} at lower at lower position than the Fe-rich spinel where Co is diffusely distributed. Although that paper employed a different nomenclature of the bands and exact Raman positions are not reported, it is clearly visible that the spectra of C-rich spinel exhibits the $T_{2g}(2)$ band in 450 cm^{-1} region somewhat. However, in most of the spectra the band had a single Lorentzian shape peaking between 465 and 467 cm^{-1} , widened on the higher wavenumber side by the $T_{2g}(3)$ b band, and at the lower wavenumber side by the $E_g(2)$ mode. Proper deconvolution of the two components is difficult due to their small separation (i.e. 20 cm^{-1} in $T_{2g}(2)$ band vs. 80 cm^{-1} separation in A_{1g} band). The width of the $T_{2g}(2)$ band ranged between 51.00 and 63.97 cm^{-1} , however, a trend across the annealing treatments was not apparent, due to its complex nature. Since a gradual increase of Fe^{2+} occupancy of A-sites, it would be expected that here the Fe-O(B) would gradually weaken in the favour of strengthening Co-O(B) mode.

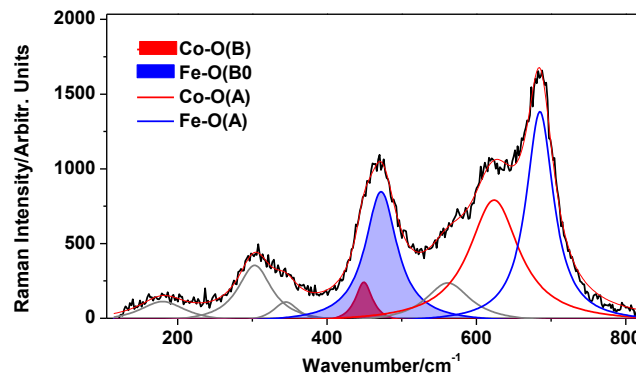


Figure 4. Example of a Co-ferrite (sample CoF-P, repetition 5) where the Fe-O(B) and Co-O(B) modes could be deconvoluted from the overall $T_{2g}(2)$ envelope around 470 cm^{-1}). The respective modes are plotted with colored fill, the modes arising from A-sites are plotted without fill, while other bands are plotted in gray for reader's reference.

The $T_{2g}(3)$, centred between 562 and 568 cm^{-1} in our samples, has often been assigned incorrectly or omitted altogether because it is completely overlapped by other stronger modes. However, this band appears very consistently in $550 - 570 \text{ cm}^{-1}$ range in other types of ferrite, such as Cu-ferrite (Mazurenko et al., 2025; Subha et al., 2018), Ni-ferrite (Cherpin et al., 2021; Duraia et al., 2019; Kamble et al., 2015; Palacio Gómez et al., 2020) and Zn-ferrite (Lazarević et al., 2014; Milanovic et al., 2016; Milutinović et al., 2013). Finally, $T_{2g}(3)$ is clearly enhanced in polarized Raman measurements of magnetite (Shebanova and Lazor, 2003), and a theoretically predicted in Co-ferrite

(Almutairi, 2025), indicating that this band is not a deconvolution artefact. The band is theoretically assigned to the combination of all the bending modes, including A-O-B, O-A-O and B-O-B (Almutairi, 2025).

The E_g band arises from a complex combination of stretching and bending modes with participation of both A- and B-sites (Almutari) exhibited a two-component character, with the stronger mode termed here $E_g(a)$ exhibited certain shift across the annealing treatment, shifting from 305 cm^{-1} in CoF-P to 301 cm^{-1} in CoF-600. Likewise, the weaker mode, termed here $E_g(b)$ exhibited a shift from 352 cm^{-1} to 347 cm^{-1} in CoF-600. This E_g appears as a single mode in bulk magnetite containing only Fe cations, however, shoulders are activated in the lattice defect containing maghemite (Chourpa et al., 2005). A shoulder to the main E_g component was also reported in Cu-ferrite with similar position and slightly smaller FWHM (Mazurenko et al., 2025). Lattice defects and sublattices that do not notably distort the atomic planes are difficult to evidence by XRD patterns (Jørgensen et al., 2007), however, may manifest in Raman spectra which are more sensitive to this kind of structural modifications. Hence, it is tempting to assign the $E_g(b)$ shape to distortion of the ferrite lattice symmetry caused by the presence of Co^{2+} , more so since the shape was previously also revealed in deconvolution results of Co-ferrite excited under 785 nm laser (Thanh et al., 2024). The mutual intensity ratio of $E_g(a)$ and $E_g(b)$ was 3.94 ± 0.50 and did not change notably across the treatments. However, their positions seem to exhibit a red shift to lower wavenumbers with greater annealing temperature (Figure 5), while their width increased. Analysis of single point spectra showed that these shifts of E_g components properties is not an artefact of spectra averaging. Nevertheless, we advise caution in analysis of position and width of these modes since $E_g(b)$ and significantly overlapped, and thus prone to deconvolution imperfections and rounding artefacts.

The $T_{2g}(1)$ band is assigned to the translational vibration of the entire metal- O_4 tetrahedron. In our case, this band was positioned between 179 and 182 cm^{-1} . This band is rarely discussed in detail in the literature, presumably because of its low intensity and proximity to the filter cut-off.

Laser wavelength effects

The band configuration of all the Co-ferrite samples was notably different under 632.8 and 830 nm excitation compared to 532 nm in terms of band configuration (Figure 2). The Fe-O(A), the Co-O(A) and the $E_g(a)$ mode was blue-shifted in 632.8 nm spectra, but red-shifted in 830 nm. The $T_{2g}(2)$ band exhibited blue shift in both cases. The $T_{2g}(1)$ mode did not show meaningful shifts. Band shifts between the two laser lines in this context could be ascribed to excitation wavelengths effect and do not necessarily hold diagnostic relevance. Comparison of band width is difficult here, as the bands are significantly overlapped. From the spectroscopy point of view, in the

absence of resonance effects, longer wavelength excitation should produce wider bands due to, among other, spectrograph grating with lower groove count (1800 gr/mm used with 532 nm laser; 1200 gr/mm used with 632.8 nm laser). Nevertheless, the intensity Co-O(A), Co-O(B), and $E_g(2)$ modes increased notably, which is an instrument-independent property and merits more detailed discussion. However, even more meaningful difference between the employed laser lines were changes in relative intensity of Co-related and Fe-related modes, which is discussed in more detail below.

The Co-O(A)-to-Fe-O(A) intensity ratio, on average across annealing treatments increased from 0.49 ± 0.54 to 1.01 ± 0.11 and 1.83 ± 0.65 passing from the 532 to 632.8 and 830 nm excitation, respectively. In terms of area, this increase was from 1.09 ± 0.08 (532 nm) to 2.71 ± 0.31 (632.8 nm) and 3.46 ± 1.72 (830 nm). These ratios do not have immediate structural significance, however, they do quantitatively illustrate the greater scattering cross-section of A-sites occupied by Co^{2+} under longer wavelength lasers. Examining each annealing step separately, we note that the 632.8 nm laser also shows a trend of increasing relative area of the Fe-O(A) after higher temperature exposure. Hence, it follows that the cation migration discussed in previous sections can also be tracked by longer wavelength lasers, although a scaling factor, which is yet undetermined, would have to be applied for quantitative calculation of X in latter case. Figure X shows comparatively the Co-O(A)/Fe-O(A) intensity and area ratios for the green and the red laser (Figure SX displays the graph with the 830 nm data included). Data obtained with the 830 nm laser are not shown here because they are too inconsistent and thus this excitation may not be appropriate for this purpose..

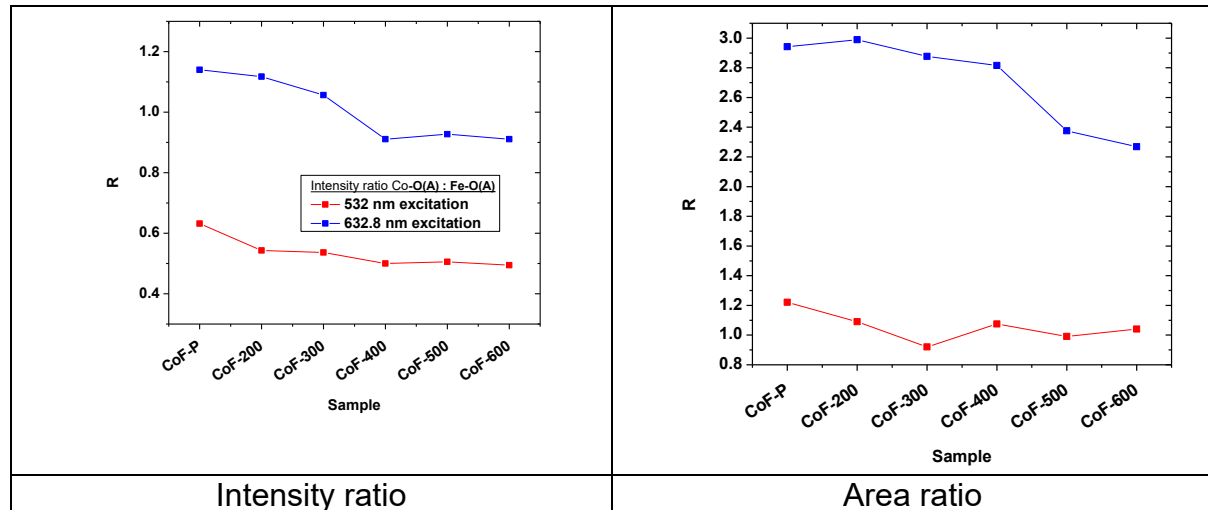


Figure 5. Intensity (a) and area (b) ratio of the A_{1g} component modes $R = I_{\text{Co-O(A)}} / I_{\text{Fe-O(A)}}$ and $R = A_{\text{Co-O(A)}} / A_{\text{Fe-O(A)}}$, respectively, in Raman spectra of Co-ferrite, acquired under the 532 and the 632.8 nm excitation. The plots highlight that cation migration upon annealing treatments is seen under either excitation line through evolution of the band intensity and area ratio.

The $T_{2g}(2)$ featured a near-perfect Lorentzian shape also under 632.8 and 830 nm laser lines. The band was significantly enhanced under the latter laser lines

compared to Fe-O(A) mode, with their intensity ratio ($R = I_{T_{2g}(2)} / I_{Fe-O(A)}$) increasing from 0.71 ± 0.02 (532 nm) to 1.74 ± 0.08 (632.8 nm) and 2.34 ± 0.65 (830 nm). The width was slightly lower under 632.8 and 830 nm excitation (49.30 ± 2.68 and 50.15 ± 3.39 cm^{-1} , respectively) than under 532 nm (52.39 ± 5.01 cm^{-1}). Here the explanation could be that the separation of Co-O(B) and Fe-O(B) is even smaller, or that one of the components was significantly enhanced (similarly as in the A_{1g}) and thus completely engulfing the other.

The $T_{2g}(1)$ band was split under 632.8 and 830 nm excitation into two components in the pristine sample and at all annealing temperatures. Similar split was observed earlier by Lazarevic et al. (Lazarevic et al., 2024), but under 532 nm excitation where the lower wavenumber component was assigned to the heavier Co-O₄ lattice vibration and the other component to the lighter Fe-O lattice. In our case, the narrower component was recorded between 126 and 129 under 632.8 nm and somewhat higher under 830 nm excitation, between 126 and 132 cm^{-1} . The broader component between 176 and 178 cm^{-1} under 632.8 nm and between 176 and 187 cm^{-1} . Considering similarities between the two excitation lines, it would rather appear that the $T_{2g}(1)a$ is preserved, while $T_{2g}(1)b$ is activated under 632.8 nm excitation, and its intensity seems to increase relative to $T_{2g}(1)b$ with greater annealing temperature. Furthermore, the $T_{2g}(1)b$ exhibits a temperature-dependent blue shift, while the $T_{3g}(1)a$ does not evidently show such a trend (Figure 7). This behaviour of $T_{3g}(1)a$ is potentially linked to the increase of Fe-O(A) intensity and area relative to the Co-O(A) mode at higher annealing temperatures, indicating slightly greater amount of Fe-O₄ tetrahedra. Taking into account the consistency of behaviour of $T_{2g}(2)$ complex with the A_{1g} modes, the assignment of $T_{2g}(1)b$ to the cobalt sublattice seems to be indeed supported.

The E_g band exhibited the complex structure also under 532 nm excitation however, the intensity ratio differs. The $E_g(a) : E_g(b)$ ratio was about 3.94 ± 0.50 in 532 nm spectra, however, it fell to about 2.45 ± 0.31 in 632.8 nm spectra. Under the 830 nm laser this band was not observed, except for slight increase of the signal around 300 cm^{-1} (not stronger than the $T_{2g}(1)b$). The native values indicate that this is due to enhancement of $E_g(a)$ rather than attenuation of $E_g(b)$ under the red laser. Considering the overall enhancement of cobalt sublattice discussed hitherto, it is tempting to correlate the $E_g(b)$ mode to vibrations involving Co cations. Unfortunately, we have no data to support this assignment, hence for the moment we just report our observations on this issue.

Spectra acquired under the 442 nm line featured a considerably different bands configuration, which cannot be interpreted in the same framework as the above spectra recorded under the previous three laser lines. Nevertheless, the spectral shape and the deconvoluted modes were consistent in all samples, from CoF-P through CoF-600. We consider the laser-induced phase transitions under this specific wavelength radiation (442 nm) after annealing at up to 600 °C unlikely. All spectra consistently exhibited a sharp mode at 718 cm^{-1} . Admittedly, maghemite exhibits a band at this position (Chourpa et al., 2005), however, given the overall intensity and

narrow character, this is not probable. A medium-intensity broad mode at 625 cm^{-1} exhibited comparable position and FWHM to Co-O(A) mode observed under visible and NIR excitation lines. Three medium-intensity modes beyond 750 cm^{-1} were also consistently observed in all spectra. In the absence of appropriate literature reference, we tentatively assign the latter modes to phonons that are activated under UV excitation, i.e. 442 nm line.

4. DISCUSSION

RS is a versatile tool owing to the freedom in adjusting the measurement parameters tailored to the end-goal of the analysis. Therefore, exploring how changing of a parameter, in this case the laser wavelength, influences the obtainable type and volume of structural information about ferrites. To the best of our knowledge, difference in performances between the 532 and 632.8 lines for structural probing of ferrites was not reported before at this level of depth and robustness.

The technical aspects of XRD analysis, for instance, assumes bulk measurements using several cubic millimetres of sample and resulting in a reflection pattern averaged on a huge number of particles. However, in micro-Raman, a microscope is used to guide the laser to the sample, resulting in a tiny probing area when using the 100x objective, with the diameter of 721 nm under the 532 nm and 857 nm under 632.8 nm laser line, respectively (Airy disk). It is evident that a small number of acquired spectra are prone to interpretation bias, which is a drawback of traditional materials engineering studies. The current paper also features a comprehensive cross-referencing of the RS results with existing literature reports also featuring quality spectroscopy results. Hence, we consider our dataset and conclusions to be sufficiently robust from the spectroscopy point of view.

Wavelength-dependent enhancement of a set of phonons within same samples was previously shown in ZnO nanoparticles synthesized by various methods (Nekvapil et al., 2022). In that case, passing from non-resonant 532 nm laser line to near-resonant 442 nm line caused a significant change in spectral configuration and band widths. The latter laser line does not meet the conditions for full resonance excitation of ZnO, but delivered nevertheless sufficient photon energy to enhance the LO phonons. It should be noted that both in the case of cited paper and the current study we are observing the enhancement of pre-existing phonons due to different laser energy rather than activation of forbidden modes due to breakage of crystalline symmetry.

According to data reported previously for nano Co-ferrite (REF Ghanbshani et al.), our NPs with 11 to 17 nm would be expected to exhibit the absorption maximum in 300 to 320 nm, hence the Resonance Raman effect, in its narrower sense, in either green or red range is not likely. Hence, we presume a kind of excitation wavelength effect in the case of current paper.

In the case of the current study, we have strong reasons to postulate that the 632.8 nm laser line enhanced the modes related to the cobalt sublattices. We can claim with relative certainty that this is valid for A-sites occupied by Co, i.e. as seen from behaviour of the $T_{2g}(1)$ complex and the A_{1g} modes. Regarding the B-sites, this claim is only tentative, as we were not able to comprehensively analyse the $T_{1g}(2)$ band in the significant portion of our spectra. Indeed, it was blue-shifted and slightly (but inconsistently) narrower, which can be assigned to some extent to wavelength and instrumental effects, and its strength was enhanced by about 2.25 times. Furthermore, A Flores et al. (A Flores et al., 2020) and Thanh et al. (Thanh et al., 2024) have reported Raman spectra of Co-ferrites excited under 785 nm laser line, a wavelength not specifically used here. Raman spectra in both of the cited studies appear to show stronger Co-O(A), and $T_{2g}(2)$ modes relative to their Fe-O counterparts, however, the studies presented scant quantitative data.

In summary, our results allow us to deduce a trend where Co-O(A) site experiences wavelength-dependent increment of its scattering cross-section relative to the Fe-O(A) mode as the laser wavelength increases. The intensity increment seems gradual and consistent, while the area increment seems to follow a logistical trend, however, in both cases standard deviation increases significantly with longer excitation wavelength. On that note, the FWHM of Fe-O(A) and Co-O(A) was not significantly different, but the different trends arise from the fact that band area scales differently than the intensity. This trend is presumably true for Co-O components in the $T_{2g}(2)$ and $T_{2g}(1)$ bands), however, at this point we do not have enough data to support this claim.

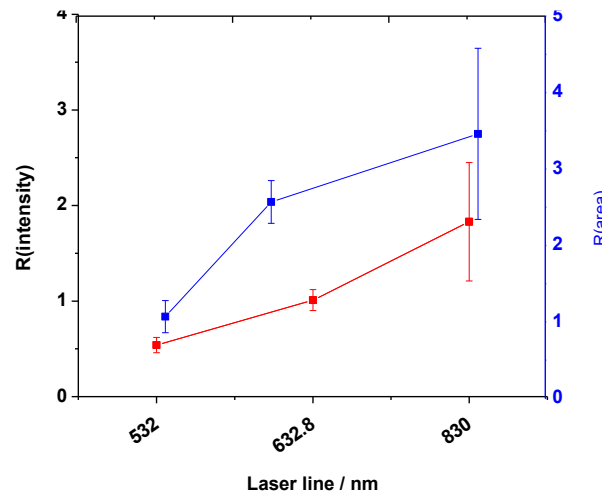


Figure 6. Intensity and area ratios plot of the A_{1g} component modes $R = I_{Co-O(A)} / I_{Fe-O(A)}$, under the laser excitations used in this study. Data points are averages of all the 6 annealing treatments.

Determination of the inversion degree at A-sites under 532 nm excitation is generally considered as valid. However, the results presented above show that applying the same area ratio formula to spectra recorded under longer wavelength

excitation would yield an over-estimation of Co-related modes and thus a skewed result. We identify a need for a targeted study which would determine the scaling factor to be applied to the area ratio formula that would account for the variation of Co-related scattering efficiency across different laser lines.

5. CONCLUSION

We have prospected the performance and type of information obtainable from Co-ferrite NPs by the 532 nm laserline which is considered a standard in Co-ferrite characterization, and the 632.8 nm laser line, which is occasionally used. The spectrum configuration is notably, but not fundamentally different between the two laser lines. Namely, the 632.8 nm laser lines enhanced the vibrations of the Co-related sublattice relative to the 532 nm line. Specific findings are as follows:

- Calculation of the inversion degree at A-sites via the area ratio of the A_{1g} components under 532 nm excitation is widely accepted; however, this calculation requires a scaling factor if the spectra are recorded under 632.8 nm excitation line.
- The Co-O(A) mode is preferentially enhanced under the 632.8 nm and 830 nm laser lines, potentially allowing targeted studies of the A- sites occupied by Co^{2+} ,
- under 632.8 and 830 nm excitation, the $T_{2g}(2)$ band is enhanced relative to Fe-O(A), however, we could not determine here if the Co-O(B) was enhanced relative to Fe-O(B),
- under 632.8 nm excitation, the $T_{2g}(1)$ modes becomes split, or rather, the cca. 124 cm^{-1} mode becomes active. We found strong support that this mode, termed here $T_{2g}(1)b$, is related to the Co-O₄ vibration.

This study brings to light two issues: (i) the 632.8 nm laser line allows preferential enhancement of Co-sublattice, (ii) the 830 nm excitation is not appropriate for quality control of Co-ferrite NPs as its spectra exhibited high deviation margins.

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